# Five trends in digital health and what they mean

Digitalization is supposed to save the healthcare system – but benefits, risks, and social consequences remain open. Where is the journey heading?

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A combination of digital solutions is intended to play a central role in the reform of the healthcare system. While expectations are high, reality shows how slowly implementation processes often proceed in practice. Digitalization and Artificial intelligence – which must be clearly distinguished from genuine intelligence – are not self-starters, as they have an impact on patients, healthcare providers, and their relationships with each other. An analysis of five developments.

## 1. The ePA as the future app for everyone and everything

About a year after its rollout, feedback on the electronic patient record for all (ePA) is mixed: while there are reports that the ePA has helped in specific treatment cases, for example, because the medications taken were visible. However, overall, complaints from healthcare providers predominate. Usability is too low, medical information was often still missing. This is also because many hospitals are far from being connected to the ePA. User numbers from insured persons are negligible, which is why many users continue to struggle with the electronic patient record. Only about five percent of statutory health insurance members currently have the technical possibility to access their ePA app at all.

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In an interim assessment, consumer protection advocates see insufficient benefit for patients and call for prioritization of elements that offer real added value, such as the digital vaccination certificate.

Many people do not even know yet that the ePA already exists and that sensitive medical information can be retrieved from it. The German AOG Foundation has already received reports from patients where their HIV status was revealed from the app, leading to uncomfortable inquiries and cancelled appointments.

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So, while reality shows that initial treatment benefits are hard won and not all people benefit equally from an ePA, politics is already moving forward. When Health Minister Nina Warken presented an update on the digitalization strategy in healthcare in February, the ePA was at the forefront of the narrative. It is intended to be an app with concrete added value in the future, including digital initial assessment and appointment booking. This is also intended to drive up user numbers. According to the strategy, there should be 20 million ePA users.

Still unresolved: if the ePA is to be continuously expanded into the central application for everything – what happens to those insured individuals who have objected to it?

## 2. Patient management and system access using AI

In the context of a future-primary care system, there is much practical discussion about digital initial assessment. This is intended to better manage how people access the right place in the healthcare system. Unnecessary visits to specialists are to be avoided, for example. For private health insurers, data and AI are also the key to more efficient management of patient flows.

Behind the scenes, there is a struggle over what the technical basis should be – and who is responsible for its implementation. This also includes whether to rely on a rule-based system or an LLM-based solution. In a position paper, North Rhine-Westphalia's Minister of Health, Karl-Josef Laumann, describes the "ideal Case" of "intelligent initial assessment software" that "complements" a standardized system with artificial intelligence.

Laumann specifically names "Structured Medical Initial Assessment" (SMIA). Patients enter symptoms there, are asked about the severity of their symptoms, and often have to answer further health questions – for example, whether they have cancer. Based on this, the urgency of the matter is then classified, from "a few days of wait" to a recommendation to go to the emergency room.

The decision is made based on filter questions and algorithmically, meaning it is reproducible at least with the same input from users. However, it remains to be seen how medically helpful such a system will actually be if one of the main concerns of users is to get an appointment with a specialist as soon as possible and to make corresponding entries. To avoid misuse, a documentation of the initial assessment is also apparently planned. The ePA is also being considered for this, which brings back the question of how to deal with ePA objections in the future.

Meanwhile, the possible role of artificial intelligence in initial assessment is completely nebulous. Studies, at least, repeatedly show how poorly language models perform in patient triage so far: clear emergencies are by no means always recognized as such. Results depend heavily on user input and are therefore not reproducible. How well a language model responds also depends, not least, on the digital and linguistic competencies of users and could therefore exacerbate social inequalities.

It remains to be seen whether the introduction of the system will actually bring the promised savings and relief to the system. Other instruments, such as telephone sick notes, which already keep people with acute infections away from crowded waiting rooms without any AI, are politically on the brink.

## 3. Automation of documentation through voice recording

If the digitalization strategy has its way, AI-supported documentation should arrive in the examination room as quickly as possible. 70 percent of the facilities are to actively use it by 2028.

The idea: Conversations in the examination room are recorded, an AI summarizes the conversation, structures it, and saves healthcare providers the manual documentation.

While at least speech recognition and transcription would be possible even with locally hosted models, reality is predictably heading in a different direction. Providers in German-speaking countries such as DocGlide or Jameda have Gemini, OpenAI, and Co. in their privacy policies for their documentation solutions. This is not ruled out, the BMG explains to 52 Dossier. The priority is to provide relief "promptly." This de facto means digital dependence on US providers will increase even further, including in healthcare.

Ideally, insured people benefit from healthcare providers being less distracted by documentation and more engaged in the conversation. However, the hurdles to discussing sensitive topics such as mental health problems or sexually transmitted diseases could increase if an Alexa-like device is present, recording everything said and sending it to the cloud.

For doctors and therapists, on the other hand, documentation is often more than an end in itself. It can also represent a final reflection on a case or help in processing cases. Does automation ultimately bring more time for patients, or a more intensified work routine because more people are to be treated in the same amount of time? The demographic change and cost pressure suggest the latter.

## 4. Artificial intelligence in treatment

Automation in documentation is just one of the examples where AI has long since arrived in care. Its use is based on context, which in practice is likely often handled by signing a data protection form during the first visit to a facility.

At the same time, it is becoming increasingly difficult for patients to foresee in what forms artificial intelligence is being used. For many people, it makes a significant difference in their own assessment whether the use relates to writing doctor's letters, the surgical robot, or radiological reporting. But how much choice do patients really have when they have to decide on a facility or a specific treatment?

Dealing with "AI" functions can also be challenging when statutory health insurance does not cover the service. For radiological findings, the offer of a second opinion via AI has long been part of the repertoire of Gaf. (individual health services) offered by many practices.

However, cost coverage is still pending in most cases, although due to the strength of machine learning systems in recognizing patterns, there is a clear advantage over purely human evaluation of findings. We are therefore moving towards a healthcare system that could become even more dependent in the future on what one can afford.

## 5. Self-responsibility in healthcare and AI

While social media and online shops are flooded with health offers, it is becoming increasingly difficult for many people to distinguish what can be sensible and what cannot. disinformation is increasing.

A stronger orientation towards health promoting offers and prevention would be sensible. Health literacy is more important than ever in the face of an unmanageable amount of misinformation. And technology is also supposed to help here.

"AI can provide citizens with personalized health recommendations in the future," the digitalization strategy states. AI-supported self-treatment is intended to relieve the system. It is intended to "strengthen self-responsible health management as a constant companion and answer questions about diseases."

While this can be successful in individual cases, and there are now reports where language models have helped in the treatment of difficult-to-diagnose diseases, it is entirely clear that technology and self-responsibility alone are not sufficient as a generalized solution.

Because whether a language model offers sensible health advice or sends people in completely wrong directions again depends on the input they use to prompt the system. Those who are already well-positioned with their digital skills benefit the most. Therefore, with a view to social inequalities, a massive expansion of educational offerings and structural prevention measures that act beyond individual health decisions would be even more important.

## Digital developments require more social support

The developments in digital health are therefore ambivalent – and not all of them are good or bad. But: The healthcare system is a social system, and new technologies have different and often unexpected effects on both healthcare providers and patients.

Especially in times of major reforms and technical adjustments, there is therefore a need for new social services that provide orientation and low-threshold assistance with health topics. Otherwise, the supposed "savior" artificial intelligence threatens to further exacerbate social inequalities in healthcare.

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